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System performance is everything, and no one is ever content with what they have. For desktops especially, gaming has always been the driving force. In most cases, upgrades are necessary, but do you absolutely need to upgrade all the components in your PC?

Do you need to upgrade?

Most desktop users using their PC for office applications, browsing and email don't absolutely need a faster PC. Most of the time, fragmented drives, bloated software and malicious soft-

ware bog down the the entire system. Upgrades will speed up your PC, but the benefits you'll notice will be just a second or two during load times or processing. Upgrades are recommended every two to three years at least.

that is capable of performing much better, but is limited due to another slower components. Gamers commonly like to upgrade all PC components at once. Graphics cards are often bottlenecked by slower processors (for example, if you use a five-year old processor and a fast Radeon 5970). There are certain games which are more CPU dependent while some others depend more on the GPU. If you play only those few specific games, then just a simple graphics card upgrade should be sufficient.

Similarly, if you use an old 5400 rpm IDE drive on a modern system, you're likely to see a big drop in read and write performance, which inadvertently

the same or slightly better than hard drives but the write speeds aren't still much better. If you're thinking value for money, they aren't the ideal choice. Now isn't the time to upgrade to one.

More than 4 GB RAM make sense?

Most 32-bit operating systems can only utilise around 3.25 GB of memory. 4 GB these days is the sweet spot when it comes to memory. Any more, and you should upgrade to a 64-bit operating system. Most memory intensive games will only use around 1.5 to 2 GB of memory.

More core vs more speed?

A minimum of two or three cores per processor is always desirable for performance. In applications that can fully utilize all cores on a processor, more cores makes more sense. At present, 3d modeling and rendering software are

FASTER CORE SPEED?

So is your system isn't fast enough for games? Look up the Overclocking your CPU and GPU stories in this issue, and also the Overclocking Guide in the November 2010 of Digit.

applications which can do this well. Most games do not use all cores fully. A faster core speed processor is often better performing in games. For example in some games, a 2-core processor at 3.2GHz is more likely to perform faster than a 2.4 GHz quad core. If you're just



3d modelling software are optimized to use cores to the fullest

using a single core processor, then its recommended that you upgrade to at least a three or four core one. For basic desktop applications, a simple dual core is still good enough. At the same time, a really slow core speed six-core processor isn't much good either. 

Upgrading Smartly

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What are the components to upgrade?

The processor, motherboard and RAM are components that are interdependent on each other. Very few people upgrade processors every few months. In fact, it's done every alternate year or so. A processor upgrade usually means a motherboard and memory upgrade as well. For example, if you've been using a Intel Core 2 Duo so far, you're very likely to upgrade to a Core i5 or i7, and a corresponding motherboard along with the DDR3 memory. Power supply and cabinets in comparison don't need to be upgraded very often, unless of course you upgrade to a really power hungry graphics card.

Bottlenecks - what are they?

Bottlenecks in PC hardware occur when you upgrade a particular component

will affect real-time PC performance.

Does adding a graphics card on an old system make sense?

Many graphics card manufacturers claim that adding a graphics card can speed up performance. It's partly true, but it's only possible on certain applications and the performance benefits are often negligible.

Should I upgrade to an SSD?

SSD (solid state drives) come in different sizes and speeds. The cheapest of SSDs are still several times more expensive than standard hard drives. The read speeds of most SSDs are



Upgrading to SSD might not always mean much higher disc transfer speeds